ANNA TAM LY

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PROFESSIONAL SUMMARY

MSc student in Mathematical and Statistical Sciences (Modelling, Data, and Predictions) at the University of Alberta. Through my interdisciplinary HBSc in mathematics, statistics, and quantitative biology at the University of Toronto, I developed a rigorous theoretical foundation that my MSc now extends into three applied skill sets:

- **Data Science:** Skilled in Python and R for data cleaning, analysis, and visualization; SQL and MySQL for database creation and management; Tableau for insight dashboards; and Git/GitHub for reproducible workflows. Built a computer-vision application that automated traffic-data reporting for a business improvement area.
- **Machine Learning:** Trained in supervised methods (regression, SVMs, tree-based models, neural networks) and unsupervised methods (clustering, PCA, factor analysis). Applied seven models to fill gaps in hourly soil-moisture records, validate predictions, and identify key environmental predictors for a government agricultural data tool.
- **Mathematical & Statistical Modelling:** Experienced in using MATLAB to translate real-world systems into models, analyze dynamics over time, and quantify uncertainty. Developed models of infectious-disease control and ecological adaptation for research labs specializing in artificial intelligence, public health, and genetics.

RELEVANT EXPERIENCE

AI & Business Data Analyst Intern **May 2026 - Aug 2026**

Old Strathcona Business Association, Edmonton, AB.

- **AI Model Integration & App Development:** Built a Python/PySide6 desktop application using YOLO and OpenCV to detect, track, and count pedestrians and vehicles across 715 recordings (190 camera-hours).
- **ETL Pipelines & Database Management:** Developed a 12-camera traffic-counting workflow covering video preparation, counting-line configuration, and SQLite data storage for a 600-business district.
- **Data Validation & Quality Assurance:** Created confidence-based review tools and manual audits to flag detection errors, improving enter/exit accuracy from 51% to 89% against 1,263 manually counted crossings.
- **Data Analytics & Reporting:** Generated hourly, daily, camera-comparison, weather-enriched, and event-level analyses, visualizations, and reports from 50,000+ accepted pedestrian counts across three event dates.
- **Product Iteration & Technical Documentation:** Translated staff feedback into product improvements and produced user guides and quick-start documentation for ongoing staff use.

Data Science Graduate Student Researcher **May 2026 - Aug 2026**

University of Alberta, in collaboration with Alberta Ministry of Agriculture, Edmonton, AB.

- **Time-Series Data Preparation:** Built Python/Jupyter Notebook workflows to prepare raw Alberta Climate Information Service (ACIS) data, covering ~900,000 hourly records × four soil-depth variables across five stations.
- **Data Imputation & Predictive Modelling:** Developed a three-level gap-classification strategy and applied Random Forest, XGBoost, LightGBM, CatBoost, LSTM, GRU, & TCN to reduce missing rate from 3.7% to 0.02%.
- **Model Validation & Evaluation:** Simulated missing-data gaps by masking known observations, then validated predictions using MAE, RMSE, R^2, and bias, achieving $R^2 \geq 0.97$ for short-duration gaps.
- **Feature Importance Analysis:** Used permutation importance to determine which soil-moisture history, climate, and seasonal predictors contributed most to prediction accuracy across the four soil depths.
- **Research Communication:** Presented project methods, results, and limitations to university researchers and soil scientists, translating complex modelling concepts for audiences with different technical backgrounds.

TECHNICAL SKILLS

Programming Languages: Python (NumPy, pandas, SciPy, Matplotlib, seaborn, statsmodels), R (tidyverse, easystats), MATLAB, SQL

Machine Learning & AI: scikit-learn, TensorFlow, PyTorch; regression, support vector machines, tree-based methods, ensemble learning (bagging, Random Forest, gradient boosting), neural networks (convolutional, recurrent), clustering (hierarchical, k-means), PCA, factor analysis, computer vision, and NLP/LLM applications (retrieval-augmented generation (RAG), text embeddings, semantic search)

Databases & Data Tools: MySQL, SQLite, Microsoft Access, Microsoft Excel, Stata, Tableau

Mathematical & Statistical Modelling: Probability and statistical inference, generalized linear models (GLMs), Bayesian modelling (PyMC), differential-equation and stochastic modelling, simulation and Monte Carlo methods, optimization (convex and nonlinear optimization; iterative methods), model validation

Development & Workflow Tools: Git/GitHub, Jupyter Notebook, Streamlit, PySide6

RESEARCH EXPERIENCE

Infectious Disease Modelling Undergraduate Student Researcher **Sep 2024 - Apr 2025**

AI & Mathematical Modelling (AIMM) Lab, Dalla Lana School of Public Health, University of Toronto, Toronto, ON

- **Compartmental Modelling:** Built and analyzed an ODE compartmental model of a monkeypox outbreak to examine disease transmission and evaluate potential control strategies.
- **Literature Synthesis:** Reviewed infectious-disease modelling literature to inform the project's research approach, assumptions, and interpretation of results.

Spatial Ecology Research Assistant **May 2024 - Aug 2024**

Fortin Lab, Department of Ecology and Evolutionary Biology, University of Toronto, Toronto, ON

- **Data Entry:** Digitized 1,000+ historical tree records into Microsoft Excel for spatial ecology analyses.
- **Database Management:** Maintained a Microsoft Access database, ensuring data completeness and consistency.

Quantitative Genetics Undergraduate Student Researcher **May 2023 - Aug 2023**

Osmond Lab, Department of Ecology and Evolutionary Biology, University of Toronto, Toronto, ON

- **Mathematical Modelling:** Investigated a PDE model describing adaptation to a moving environmental optimum.
- **Mathematical Analysis:** Derived expressions for mean fitness and the critical rate of environmental change.

Evolutionary Genetics Research Assistant **May 2022 - Nov 2022**

Agrawal Lab, Department of Ecology and Evolutionary Biology, University of Toronto, Toronto, ON

- **Image Analysis & Data Collection:** Collected growth measurements for 200+ plants using ImageJ and Excel.
- **Statistical Analysis:** Performed t-tests to assess fitness differences across reproductive modes.

ADDITIONAL EXPERIENCE

Retail Specialist **Jul 2020 - Jul 2022**

Krispy Kreme Doughnut Factory, Scarborough, ON

- **Teamwork & Communication:** Trained new staff on doughnut and beverage preparation and communicated effectively with team members to keep service moving smoothly and ensure a positive customer experience.
- **Multitasking & Time Management:** Balanced register duties, order fulfillment, and customer requests simultaneously, consistently upholding speed and accuracy during high-volume shifts.
- **Attention to Detail & Reliability:** Processed transactions accurately and maintained clean, sanitary work areas in compliance with health and safety standards.

EDUCATION

University of Alberta, Master of Science (MSc) **Expected Dec 2026**

Mathematical and Statistical Sciences, Specialization in Modelling, Data, and Predictions

- GPA: 3.95/4.00

University of Toronto, Honours Bachelor of Science (HBSc) with High Distinction **Apr 2025**

Double Major in Mathematics and Quantitative Biology, Minor in Statistics

- GPA: 3.76/4.00

HONOURS AND AWARDS

Data Masters Challenge Olympiad **Apr 2026**

- Awarded second place in a 48-hour intercollegiate data science competition by analyzing 390 Olympic athlete records across 17 variables, engineering features, evaluating machine-learning models to predict medal success, and developing strategic recommendations for the United States and Canada ahead of the 2030 Winter Olympics.

Alberta Graduate Excellence Scholarship (AGES) **Feb 2025**

- Awarded $25,000 by the Government of Alberta in recognition of outstanding academic achievement and to support the recruitment of exceptional graduate students to Alberta.

Ecology & Evolutionary Biology Undergraduate Student Research Award (USRA) **May 2023**

- Awarded $7,500 by the University of Toronto's Department of Ecology & Evolutionary Biology to conduct a full-time research project during Summer 2023.

Dean's List Scholar **2021, 2022, 2023, 2025**

- Awarded to University of Toronto's Faculty of Arts & Science students who obtain a cumulative GPA of 3.50 or higher after completing their fifth, tenth, fifteenth, or twentieth degree credits.